



Course Project

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Logistics on fragile/sensitive items

FRAGILE

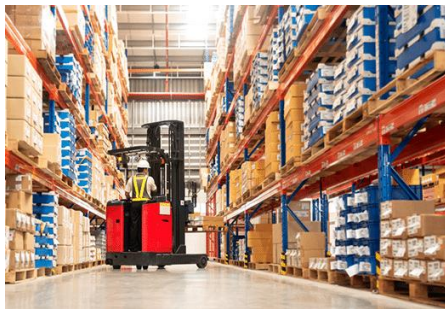




A long way to the final consumer



Production



Storage



Transportation



Delivery

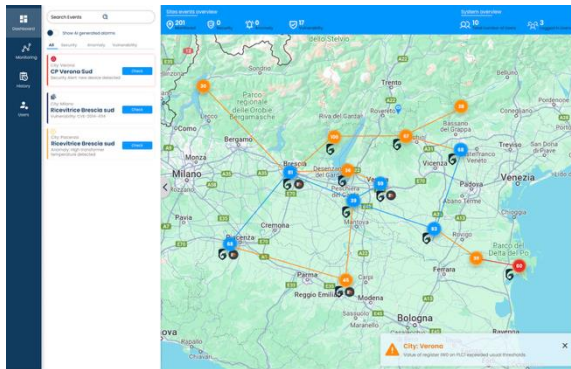
- Many providers & parties involved
- Unreliable data collection & handling: potential data tampering
- Trust problems
 - between different actors in the supply chain
 - by the final consumer



System-to-be



Securely tracking devices



Live-data dashboard



Interoperability

- Secured and certified data – sensors/data cannot be tampered with
- IoT devices + SaaS cloud infrastructure
- Dashboard for real-time data visualization and reporting
 - Processes turn transparent, traceable and measurable
- Web API to data interoperation
 - Support quality/sustainability certification of products and processes
- Data notarization with blockchain smart contracts (optional)



Functional requirements

- **Tracking:**
 - Periodic recoding/publication of certain environmental conditions (e.g., temperature, humidity, vibrations)
 - Recorded data should be digitally signed
- **Devices:**
 - Can be provisioned/decommissioned, listed, associated/dissociated to shipments
 - Devices can be configured (recording/synchronization frequency, which metrics to collect)
- **Shipments:**
 - Can be created, changed, deleted, inspected
 - Devices can be added/removed
 - Can be started/stopped/paused/resumed
- **Dashboard:**
 - Data publicly available to managers, drivers, consumers, and certification authorities
 - Evidence of “out of range” conditions (alarm)

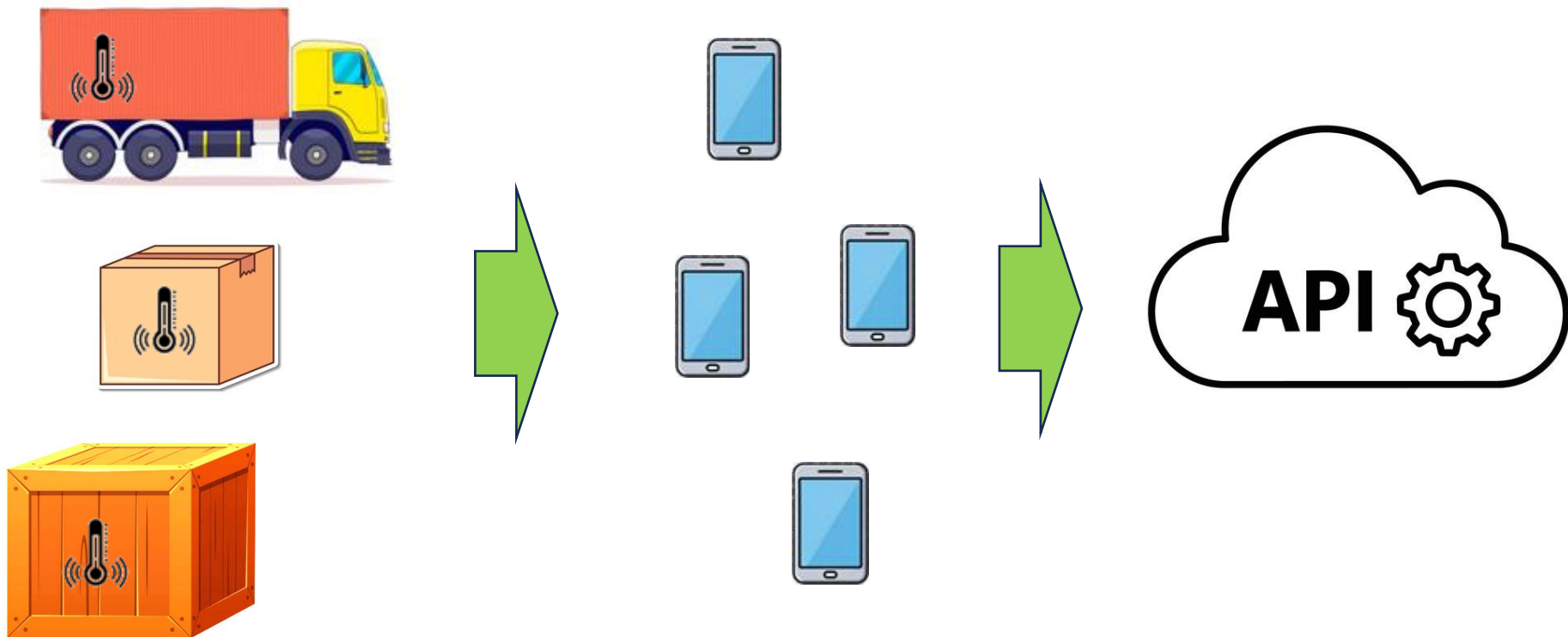


Nonfunctional requirements

- **Security and trust:**
 - Software/hardware is genuine (not tampered with)
 - No tampered/made-up data
 - Stakeholders, authorities and certification bodies can verify data
- **Low cost:**
 - Cheap commodity IoT devices
 - IoT devices are reusable for different shipments/boxes/vehicles
- **Battery operated:**
 - Low power consumption (e.g., Bluetooth low energy, no 5G, no WiFi)
 - No direct internet access
 - Battery charge monitoring, low battery notification (maintenance)
- **Scalability:**
 - Cheap server/cloud infrastructure to manage data
- **Versatile and easy to deploy:**
 - Scalable SaaS cloud infrastructure to display and interoperate data
 - Multiple configurable sensors (temperature, humidity, vibrations, UV radiations, ...)
 - Out-of-the-box operational devices (no provisioning needed)
- **Interoperability:**
 - Processes turn transparent, traceable, measurable
 - Data available via web APIs for integration (e.g., with ERP, authorities, ...)



Possible architecture





Commented goal-oriented diagrams

- Intentional view
 - Goal diagram with obstacles, obstruction links, resolution link
- Structural view
 - Class diagram for conceptual objects
- Responsibility view:
 - Agent diagram
 - Responsibility assignment
 - Agent capabilities, load analysis?
 - Dependency diagram (for vulnerability analysis)?
 - Context diagram?
- Functional view
 - System operations (operationalization diagram)
- Behavioral view:
 - State machine or sequence diagram



Requirement document: IEEE std 830-1998

- 1. Introduction
 - 1.1 RD purpose
 - 1.2 Product scope
 - 1.3 Definitions, acronyms, abbreviations
 - 1.4 References
 - 1.5 Overview
- 2. General Description
 - 2.1 Product perspective
 - 2.2 Product functions
 - 2.3 User characteristics
 - 2.4 General constraints
 - 2.5 Assumptions & Dependencies
 - 2.6 Apportioning of requirements
- 3. Specific Requirements
 - 3.1 Functional requirements
 - 3.2 External interface reqs
 - 3.3 Performance reqs
 - 3.4 Design constraints
 - 3.5 Software quality attributes
 - 3.6 Other requirements
- Appendices
- Index



Project format

- A report describing the project
 - Commented goal-oriented diagrams
 - Requirement document
- Deadline: 7 days before the exam date
 - To be delivered using moodle (one upload per group)
 - Single zip archive: 2 markdown file (at least) + pictures + objective project
 - To be done as a group (preferably 3 participants)